The electromagnetic spectrum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electromagnetic wavelength, frequency and propagation.

SUCCESS CRITERIA - I can...

1. Use c=fλ in vacuum.
2. Order spectral regions by wavelength.
3. Distinguish electromagnetic from mechanical waves.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “electromagnetic” and “spectrum” to explain your model or result. Sentence starters are optional.

electromagnetic: involving coupled electric and magnetic fields

spectrum: a range arranged by frequency or wavelength

vacuum: space with negligible matter

01 Warm-up: bring back an earlier idea

A closed-open pipe has length 0.3 m.
Use λ=4 L for the fundamental and v=340 m/s.
Find its fundamental frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

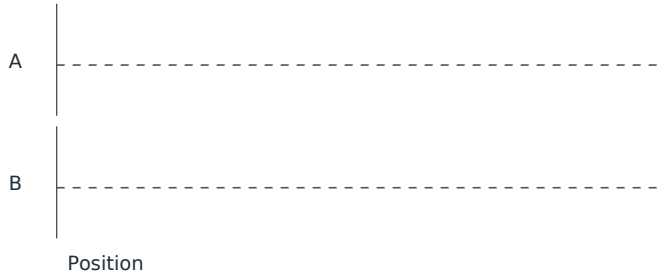
02 Worked example: explain the model

Reference: c = f λ; c = 3.0×10⁸ m/s.
Electromagnetic waves in vacuum. Frequency identifies spectral region; mechanical sound needs a medium.

Worked reasoning: In vacuum, all electromagnetic waves travel at c=3×10⁸ m/s. A frequency of 10⁸ Hz gives wavelength 3 m. Higher frequency means shorter wavelength at the same speed.

Your turn: Draw two waves with a common speed but different wavelengths; label which has higher frequency.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

03 Read and interpret the model

Order radio, visible and X-rays from longest to shortest wavelength 0. Which has greatest frequency in vacuum?

Sentence starter: I notice ... . This supports ... because ... .

Radio	IR	Visible	UV	X-ray

Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

04 Apply the physics

Compare two signals in vacuum: light of wavelength 600 nm and radio of frequency 100 MHz.
Use c = 3 × 10⁸ m/s, 1 nm = 10⁻⁹ m and 1 MHz = 10⁶ Hz.
Find the light’s frequency and radio wavelength.
Complete the table.

Sentence starter: Both ... . They differ in ... because ... .

Signal	Frequency (Hz)	Wavelength (m)
Light	____	600 × 10⁻⁹
Radio	100 × 10⁶	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why can sunlight cross space but ordinary sound from a speaker cannot cross a vacuum?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

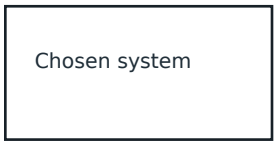
06 Review: connect an earlier idea

A heat engine operates between 600 K and 400 K and absorbs 800 J.
Find maximum ideal work and explain why actual work is usually lower.
Label energy entering, leaving and stored.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

A wave has $f = 2.4 \times 10^{9}$ Hz in vacuum.
Find λ.
Identify the spectral region.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Radio	IR	Visible	UV	X-ray

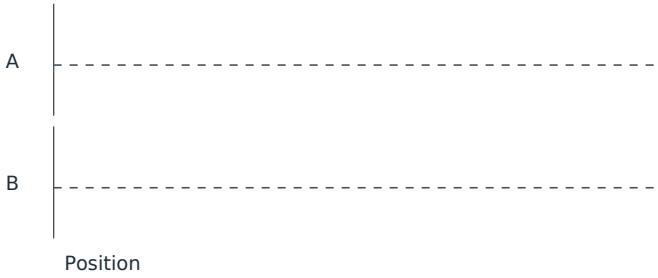
Label increasing frequency and wavelength.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Visible light has wavelength 600 nm.
Find frequency using $1\ \text{nm} = 10^{-9}$ m.
Compare with light of wavelength 300 nm.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3